FACULTY OF CHEMICAL & ENERGY ENGINEERING

SEMESTER 1 2025/2026

SETP4583 – SURFACTANT AND DETERGENT TECHNOLOGY

SECTION EB01

INDIVIDUAL REFLECTION REPORT

LECTURER: DR. AZAD ANUGERAH BIN ALI RASOL

STUDENT NAME: CHUA JIA LIN

MATRIC NO: A23CS0069

1.0 INTRODUCTION

I had limited understandings about detergents and surfactants before enrolling in the Surfactant and Detergent Technology class. I thought detergents were just simple cleaning products made by the combinations of various chemicals that are used to clean dirt in our daily lives. I expected the course to focus more on theoretical knowledge rather than practical skills, which might be challenging for me because I am not good at chemistry.

I decided to enrol in this course because I was interested in understanding how detergents such as dishwashing liquids, shampoos, and soaps are formulated and produced. My expectations for this course were to gain knowledge on how to select gentle and environmentally friendly detergent and gain a fun experience on producing detergents in lab activities. Over the 14 weeks, my expectations were met because I learned the formulation of detergent and the reason for choosing the ingredients to create a gentle and eco-friendly detergent and had a hands-on experience producing the detergent in lab sessions.

2.0 REFLECTION ON KEY LEARNINGS

One of the most important concepts I learned in this course was the classification of surfactants and their functions. I realised that surfactants are the main ingredients in detergents because they help to reduce the surface tension and allow oil and dirt to be removed from surfaces. There are four main types of surfactants, which are anionic, cationic, non-ionic, and amphoteric surfactants. Each type of surfactant has different chemical structures, properties, and applications. For example, anionic surfactants such as SLES are effective cleaning and foaming agent.

Another key concept was the role of main detergent ingredients such as SLES, LABS, NP9, CDE, Betaine, and Berol. Before learning this course, these chemicals are unfamiliar for me. However, through the lectures and lab sessions, I learned the functions and usage of these chemicals in the detergent formulation. For instance, SLES and LABS function as primary surfactants that provide strong cleaning power, while NP9 and Berol act as non-ionic surfactants that improve grease removal. The combination of these chemicals can produce effective, gentle, and eco-friendly detergents.

Other than that, the laboratory sessions were the most valuable and interesting part of this course. This session enables the application of theoretical knowledge in practical hands-on labs. During the dishwashing liquid formulation lab, I had the opportunity to apply theoretical knowledge in a practical manner. Measuring and mixing the ingredients showed me how sensitive formulations are and how small changes can affect the viscosity, foaming, and cleaning ability of the detergent. The final product made me feel confident that I had gained real technical skills to produce a functioning detergent.

Besides, individual assignments and group projects also helped me to develop deeper understandings about surfactants and detergents. For example, there is an individual assignment that request students to analyse a commercial cleaning product in different perspectives such as surfactant types and their functions, the formulation used, and the pros and cons of the product. This assignment helps me to understand a detergent in very detailed manner. Furthermore, the group projects of formulating a dishwashing liquid helps to develop teamwork and communication skills while solving formulation and testing problems together.

3.0 APPLICATION AND REAL-WORLD RELEVANCE

The knowledge gained from this course is highly relevant to real-world industrial applications because detergents and surfactants are highly used in many industries such as household cleaning products and cosmetics. Understanding the properties of surfactants such as foaming, emulsification, and biodegradability is essential for developing products that are effective, safe, and sustainable.

For example, learning about SLES and LABS helped me understand why they are widely used in dishwashing liquids and laundry detergents due to their strong cleaning abilities, while non-ionic surfactants like NP9 and Berol are commonly used in industrial cleaners and personal care products because they perform well in both hard and soft water.

The laboratory experience of producing dishwashing liquid was very meaningful because it represents real industrial processes. In a manufacturing environment, similar steps such as ingredient selection, mixing, quality control, and performance testing are followed. This experience helped me connect classroom learning to real product development. It also made me appreciate how chemical engineers and formulation chemists must consider cost, performance, safety, and environmental impact when designing detergents.

4.0 CHALLENGES AND GROWTH

One of the main challenges I faced in this course was the accuracy of weighing the ingredients. During the laboratory sessions, I realised that each ingredient in the formulation has a specific function, and incorrect proportions could affect the final product. Accurate weighing of ingredients was very challenging because small measurement errors could affect the viscosity, foaming ability, and overall performance of the dishwashing liquid. To solve this problem, I carefully weigh the amount of chemicals needed using electronic scale to prevent incorrect proportions of ingredients.

Another challenge was the need for long periods of continuous stirring during the formulation process. Maintaining consistent stirring required patience and focus, as improper mixing could result in an uneven or unstable product. This experience taught me the importance of precision, consistency, and attention to detail in detergent formulation. Working in groups during the lab sessions is helpful to solve this problem because group members can do the stirring alternately to prevent tiredness due to long period of stirring.

Through these challenges, I became more confident in handling formulation procedures. Besides, I realised that teamworking is very important because a good teamworking can increase efficiency.

5.0 PERSONAL AND PROFESSIONAL DEVELOPMENT

This course has influenced my perspective on formulation science and product development, even though I am a computer science student. Now, I appreciate how everyday products are created through the integration of scientific principles, experimentation, and systematic processes.

Learning about surfactants and detergents has broadened my understanding of industries such as consumer goods and chemical manufacturing, which I had not previously considered in detail.

The hands-on laboratory experience strengthened my ability to follow structured procedures, analyze results, and solve practical problems, which are skills that are also highly relevant in computer science. Working with chemical formulations improved my attention to detail and increased my confidence in handling technical tasks outside my main field of study. In the future, I am interested in learning more about environmentally friendly and biodegradable surfactants, as sustainability is an important consideration across all industries. Overall, this course has helped me develop soft skills such as problem-solving, teamwork, and critical thinking that will be valuable in both my academic studies and future career.